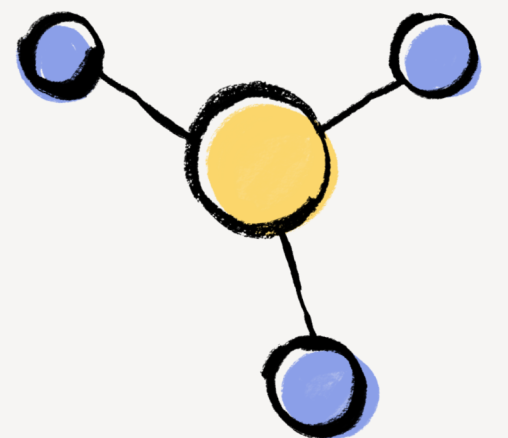
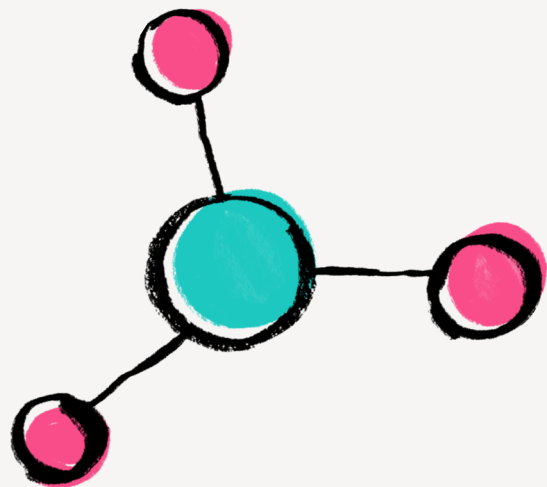
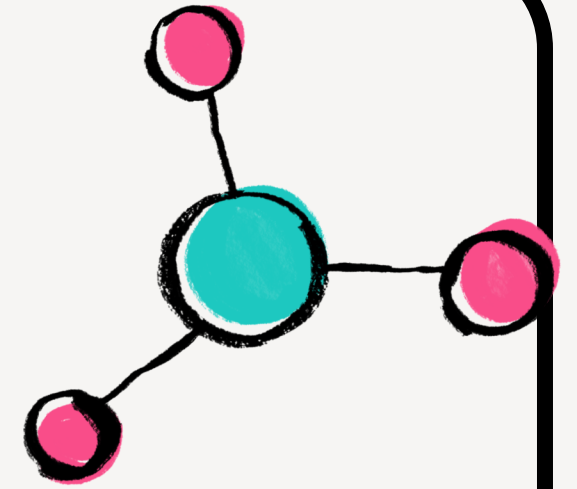


FROM SHOT TO SYSTEM : DATA DRIVEN INSIGHTS ON ALCOHOL AND THE BODY

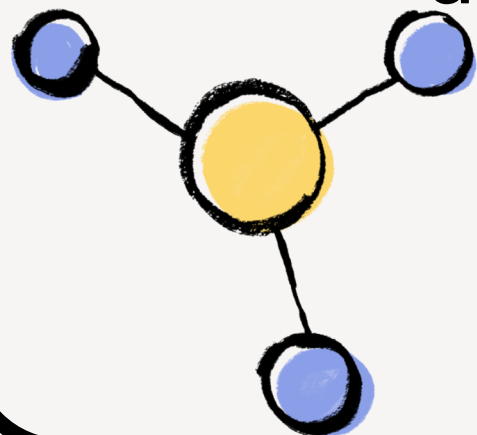
Antony Cissna, Bryan Correa, Ishika Kaushik, Sreeya Pasumarth



INTRODUCTION



- Alcohol a fan favourite drink for people from all around the world, and unfortunately a major contributor to a public health crisis.
- Data shows that the in the US alcohol related cause lead to 144,730 deaths per year.
- California leads as the state with the highest alcohol related deaths with 12,369 per year.



CLEANING THE DATASET



1. Remove unnecessary rows and columns

2. Change column names and add needed columns

3. Correct class types and clean values

4. Pivot to long data

5. Rearrange column order

6. Repeat for all tables and join datasets

Harmful Effects						
Chronic Causes	Overall	0-19	20-34	35-49	50-64	65+
100% Alcohol-Attributable						
Alcohol abuse	598	-	-	155	264	136
Alcohol cardiomyopathy	75	-	-	19	31	21
Alcohol dependence syndrome	650	-	-	163	287	158
Alcohol polyneuropathy	-	-	-	-	-	-
Alcoholic gastritis	-	-	-	-	-	-
Alcoholic liver disease	4,776	-	-	1,077	2,186	1,273
Alcoholic myopathy	-	-	-	-	-	-
Alcoholic psychosis	173	-	-	34	61	69
Alcohol-induced acute pancreatitis	74	-	-	22	26	13
Alcohol-induced chronic pancreatitis	-	-	-	-	-	-
Degeneration of nervous system due to alcohol	24	-	-	-	-	13
Fetal alcohol syndrome	-	-	-	-	-	-
Fetus and newborn affected by maternal use of alcohol	-	-	-	-	-	-
100% Alcohol-Attributable Total	6,383	-	-	1,473	2,871	1,684
Cancer						
Cancer, breast (females only)	296	-	-	29	80	184
Cancer, colorectal	299	-	-	22	77	197
Cancer, esophageal *	118	-	-	-	28	86
Cancer, laryngeal	47	-	-	-	-	36
Cancer, liver	621	-	-	16	175	427
Cancer, oral cavity and pharyngeal	283	-	-	11	75	195
Cancer, pancreatic †	79	-	-	-	17	59
Cancer, prostate (males only)	69	-	-	-	-	62

Website

usa_data					
State	Effect_Duration	Effect_Category	Effect	Age	Deaths
AL	Chronic	100% Alcohol-Attributable	Alcohol abuse	0-19	0
AL	Chronic	100% Alcohol-Attributable	Alcohol abuse	20-34	0
AL	Chronic	100% Alcohol-Attributable	Alcohol abuse	35-49	17
AL	Chronic	100% Alcohol-Attributable	Alcohol abuse	50-64	21
AL	Chronic	100% Alcohol-Attributable	Alcohol abuse	65	11
AL	Chronic	100% Alcohol-Attributable	Alcohol cardiomyopathy	0-19	0
AL	Chronic	100% Alcohol-Attributable	Alcohol cardiomyopathy	20-34	0
AL	Chronic	100% Alcohol-Attributable	Alcohol cardiomyopathy	35-49	0
AL	Chronic	100% Alcohol-Attributable	Alcohol cardiomyopathy	50-64	0
AL	Chronic	100% Alcohol-Attributable	Alcohol cardiomyopathy	65	0
AL	Chronic	100% Alcohol-Attributable	Alcohol dependence syndrome	0-19	0
AL	Chronic	100% Alcohol-Attributable	Alcohol dependence syndrome	20-34	0
AL	Chronic	100% Alcohol-Attributable	Alcohol dependence syndrome	35-49	14
AL	Chronic	100% Alcohol-Attributable	Alcohol dependence syndrome	50-64	35
AL	Chronic	100% Alcohol-Attributable	Alcohol dependence syndrome	65	16
AL	Chronic	100% Alcohol-Attributable	Alcohol polyneuropathy	0-19	0
AL	Chronic	100% Alcohol-Attributable	Alcohol polyneuropathy	20-34	0
AL	Chronic	100% Alcohol-Attributable	Alcohol polyneuropathy	35-49	0
AL	Chronic	100% Alcohol-Attributable	Alcohol polyneuropathy	50-64	0

Cleaned .csv

AGE GROUP vs EFFECT DURATION

Inference

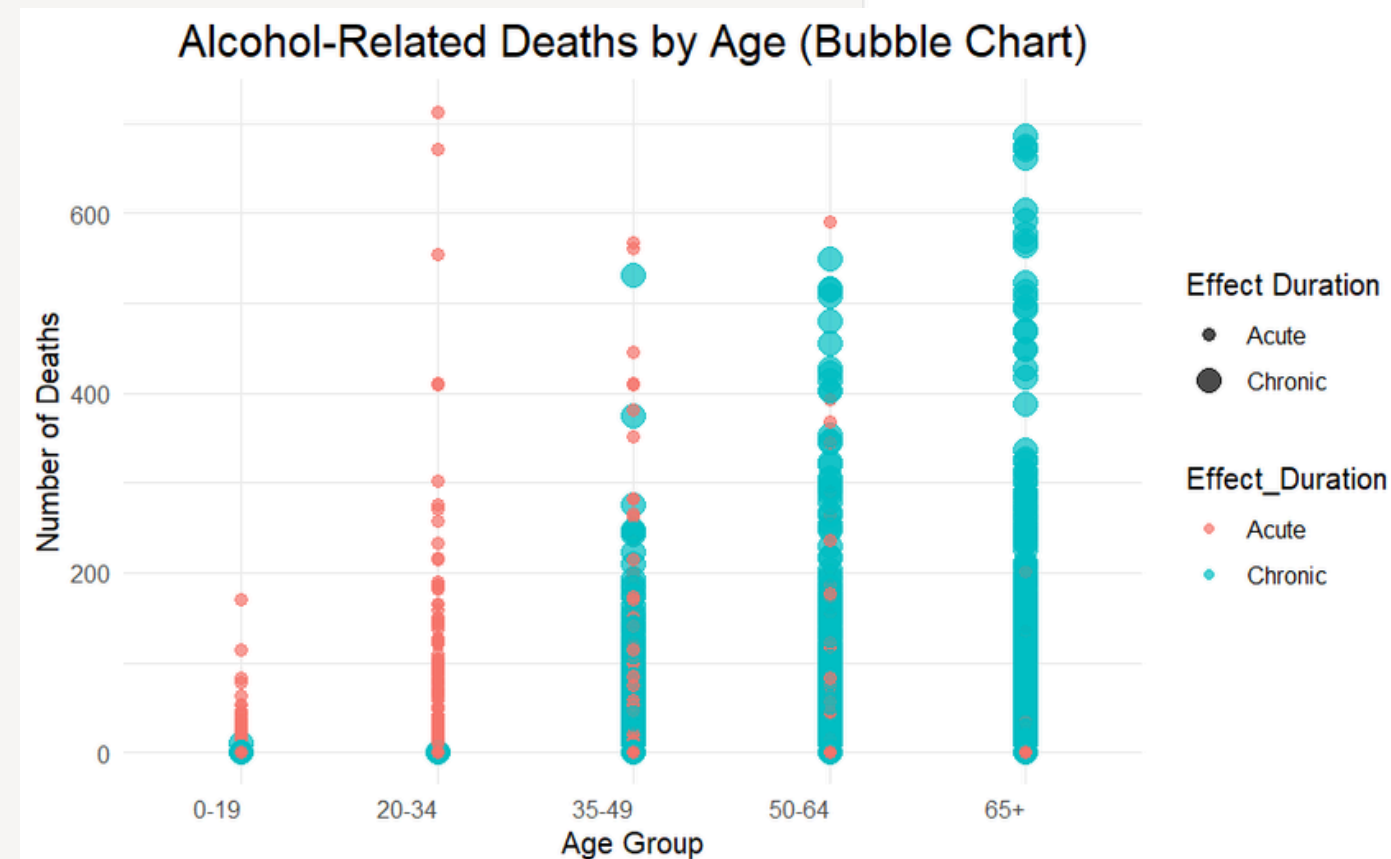
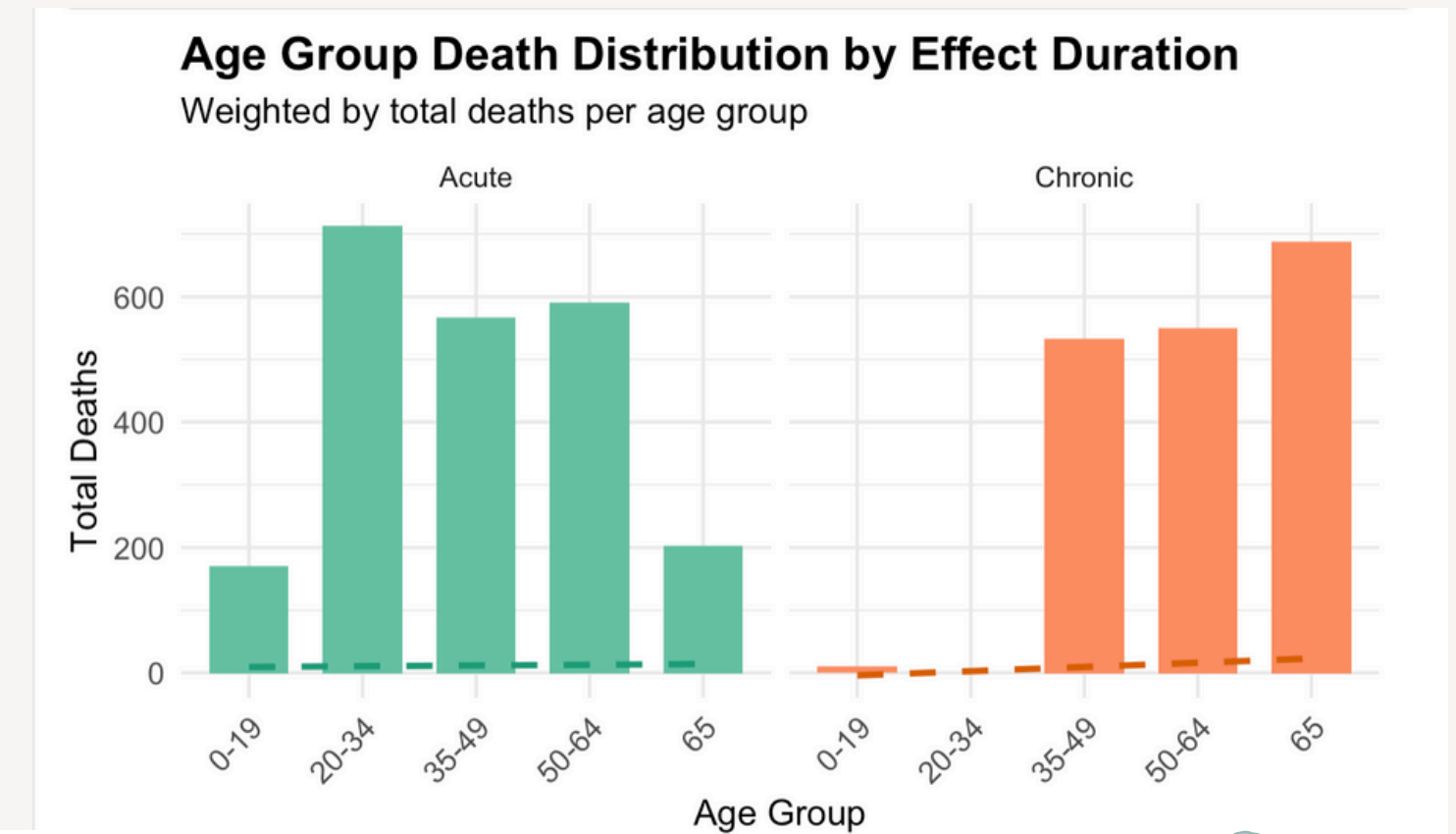
the distribution of total deaths across different age groups, separated by acute and chronic effect durations.

Observation

It highlights that acute conditions lead to higher deaths in younger adults (20-49), while chronic conditions show a sharp increase in mortality in older age groups (50+).

Future Therapies

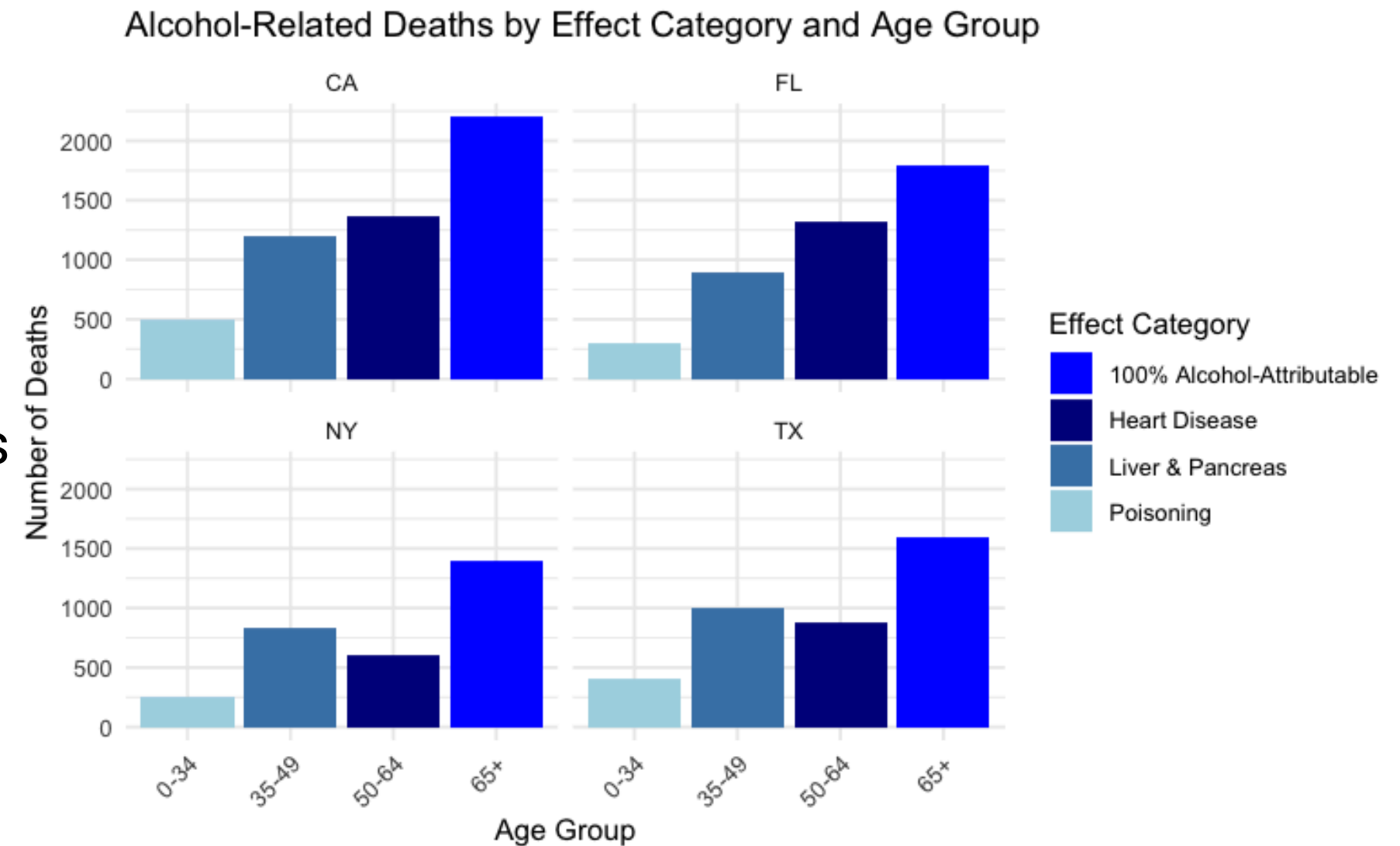
This indicates a need for age-specific interventions for both acute and chronic alcohol-attributable conditions.



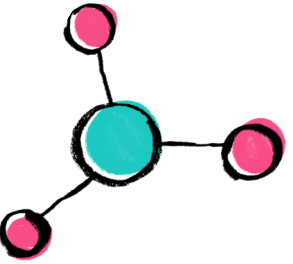


ALCOHOL ATTRIBUTED DEATHS ACROSS THE US

- We picked out 4 random states across the US, and observed the trends for alcohol related deaths.
- **CALIFORNIA:**
 - Recorded the highest number of alcohol related poisoning deaths.
- **TEXAS:**
 - Recorded the most liver and pancreas related deaths due to alcohol
- **FLORIDA:**
 - Recorded the highest number of 100% alcohol attributable deaths.
- **NEW YORK:**
 - Recorded the lowest numbers across most effect categories

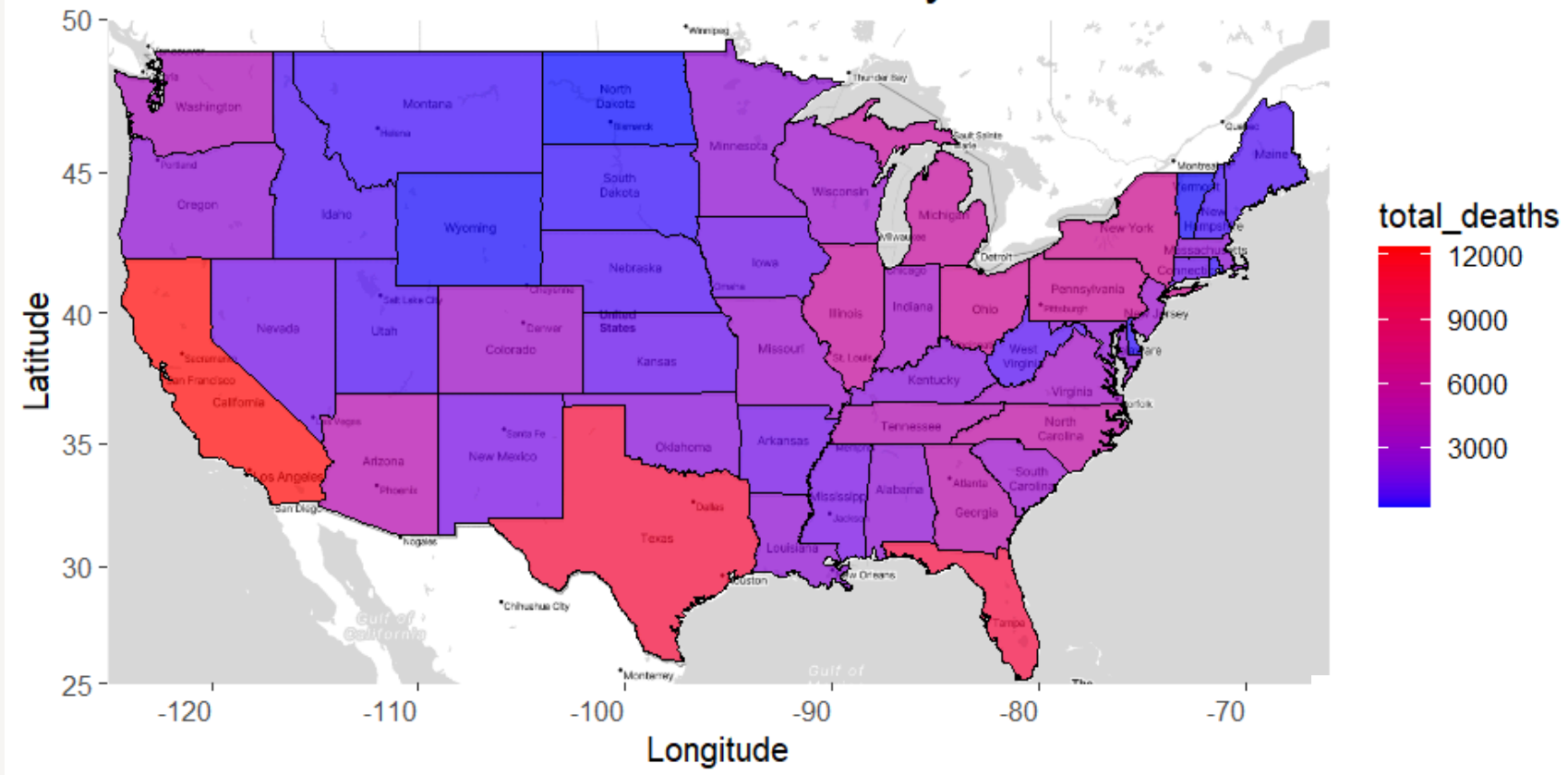


The next question is: What could be the reason for the variation in alcohol-related deaths across states?



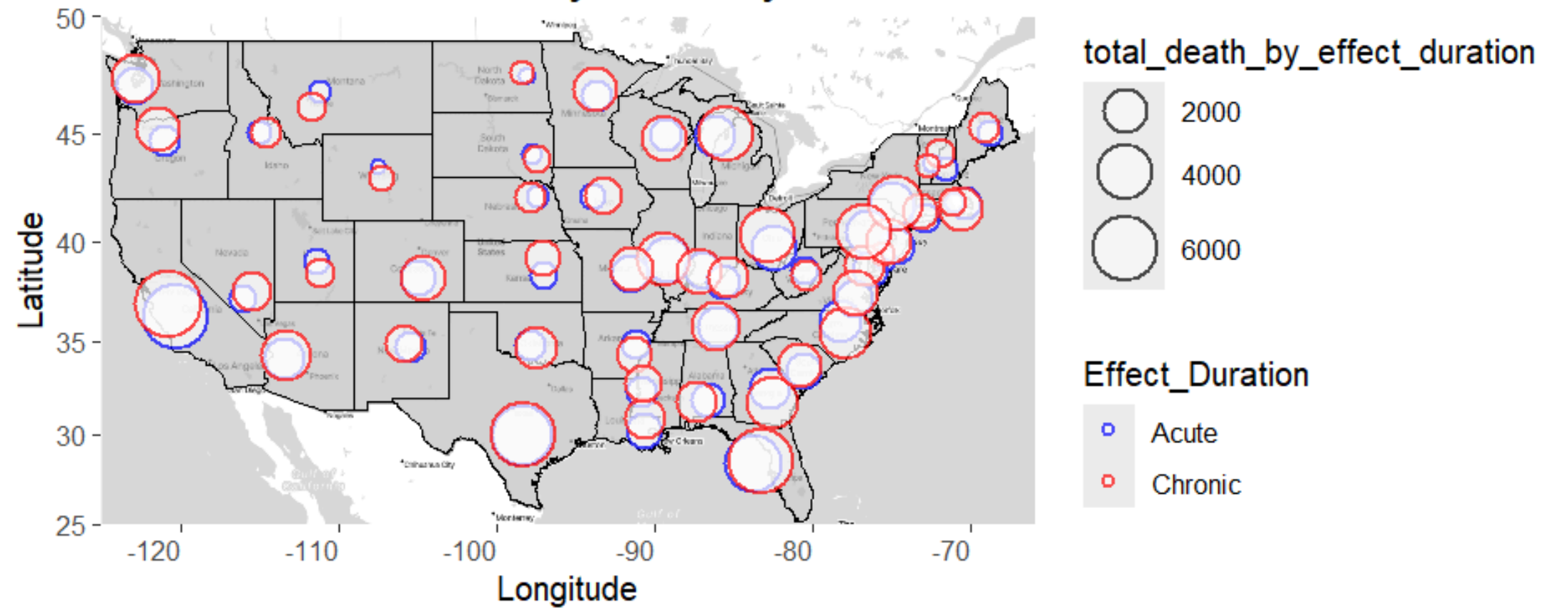
- **California**
 - Recent data has indicated an increase in substance misuse, thus leading to a rise in fatalities.
- **Texas**
 - Access to preventative healthcare in remote regions.
- **Florida**
 - Tourism and leisure culture.
- **New York**
 - Interesting case: public transportation system reduces the chances of driving fatalities under the influence of alcohol.

Alcohol-Related Deaths by State



stamen_toner_lite
geom_polygon
geom_point

Alcohol-Related Deaths by State by Effect Duration



DONT BE A STATISTIC!!



SAFE PARTY UC DAVIS

- Local Laws
- Transportation
- Substance Abuse Resources

THANK YOU

